Function Machines

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

One-Step Machine: A function machine applies the rule ×4 to its input. What is the output when the input is 7?

Answer: 7 × 4 = 28.

Two-Step (Chained) Machine: A function machine: input → +3 → ×2 → output. What is the output when the input is 5?

Answer: 5 + 3 = 8; 8 × 2 = 16.

Working Backwards: A function machine: input → ×3 → −7 → output. The output is 11. Find the input.

Answer: Reverse the steps: 11 + 7 = 18; 18 ÷ 3 = 6. The input was 6.

Function Basics (15 questions)

1. A function machine applies the rule ×5 to its input. What is the output when the input is 6?
2. A function machine applies the rule +9 to its input. What is the output when the input is 14?
3. A function machine applies the rule −6 to its input. What is the output when the input is 20?
4. A function machine applies the rule ÷4 to its input. What is the output when the input is 32? A) 6 B) 7 C) 8 D) 9 E) 10
5. A two-step function machine: input → +2 → ×3 → output. What is the output when the input is 4?
6. A two-step function machine: input → ×2 → +5 → output. What is the output when the input is 9? A) 18 B) 20 C) 21 D) 23 E) 25
7. A function machine applies the rule ×6 to its input. If the output is 42, what was the input?
8. A function machine applies the rule +15 to its input. If the output is 31, what was the input? A) 14 B) 15 C) 16 D) 17 E) 18
9. A function machine applies the rule −8 to its input. If the output is 5, what was the input?
10. A function machine applies the rule ÷5 to its input. If the output is 9, what was the input?
11. Select ALL that apply: a function machine applies the rule ×3 to each input. Which of these input→output pairs are correct? A) 4→12 B) 5→15 C) 6→20 D) 7→21 E) 8→22
12. A two-step function machine: input → ×4 → −3 → output. What is the output when the input is 5?
13. A two-step function machine: input → +7 → ÷2 → output. What is the output when the input is 9?
14. A function machine takes an input, doubles it, then adds 6. What is the output when the input is 10? A) 20 B) 22 C) 24 D) 26 E) 28
15. A function machine applies the rule −10 to its input. What is the output when the input is 10?

Working Forwards & Backwards (15 questions)

1. A function machine: input → ×3 → +8 → output. What is the output when the input is 11?

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2. A function machine: input → −5 → ×4 → output. What is the output when the input is 12? A) 24 B) 26 C) 28 D) 30 E) 32

3. A function machine: input → ÷3 → −2 → output. The input is 18. Find the output.

4. A machine takes an input, adds 4, then multiplies by 3. If the output is 33, what was the input? A) 5 B) 6 C) 7 D) 8 E) 9

5. A function machine: input → ×2 → −9 → output. The output is 15. Find the input.

6. A function machine turns 5 into 19 and 8 into 28 using the same two-step rule "×a, then +b". Find the rule (give both a and b).

7. A function machine: input → ×4 → output. What is the output when the input is 0.5?

8. Priya puts a number into a machine. The machine triples the number, then adds 5. The output is 29. What number did Priya put in? A) 6 B) 7 C) 8 D) 9 E) 10

9. A function machine: input → +7 → output. What is the output when the input is −3?

10. Select ALL that apply: a function machine applies the rule "subtract 4, then multiply by 2" to inputs. Which input→output pairs are correct? A) 6→4 B) 9→11 C) 10→12 D) 12→14 E) 15→22

11. A function machine: input → ×5 → output. What is the output when the input is 2/5?

12. A machine doubles a number and then subtracts 4. If the result is 10, what was the original number?

13. A function machine: input → ×0.5 → output. What is the output when the input is 13? A) 5.5 B) 6 C) 6.5 D) 7 E) 7.5

14. A function machine: input → ÷4 → +6 → output. What is the output when the input is −8?

15. A function machine takes a number, adds 1/4 to it, then doubles the result. What is the output when the input is 3/4?

Multi-Step Challenge (15 questions)

1. A function machine: input → ×4 → −7 → ÷3 → output. What is the output when the input is 10?

2. A function machine: input → −9 → ×3 → output. The output is −12. Find the input.

3. A two-step machine applies the rule "×a, then +b" to every input. It turns 6 into 32 and 9 into 44. What is a + b? A) 10 B) 11 C) 12 D) 13 E) 14

4. A function machine: input → ÷4 → +2.5 → output. The output is 9. Find the input.

5. A function machine: input → ×(−2) → +9 → output. What is the output when the input is 6?

6. Select ALL that apply: a function machine applies the rule "double the input, then subtract 7". Which input→output pairs are correct? A) 5→3 B) 8→9 C) 10→13 D) 12→18 E) 20→35

7. A function machine: input → +15 → ÷5 → ×2 → output. The output is 12. Find the input. A) 10 B) 12 C) 15 D) 18 E) 20

8. A function machine: input → ×7 → −20 → output. The output is 1. Find the input.

9. A function machine: input → +11 → ×(−1) → output. What is the output when the input is 4?

10. A function machine: input → ÷2 → −3.5 → output. What is the output when the input is 9?

11. A machine doubles a number, adds 9, then halves the result, giving a final answer of 14.5. What was the original number? A) 8 B) 9 C) 10 D) 11 E) 12

12. A café charges a £2 booking fee plus £3 per person for a group meal, so the total cost is found by a machine that multiplies the number of people by 3, then adds 2. What is the total cost for 9 people?

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13. A number is put into a machine that subtracts 8, then divides the result by 3, giving an output of −2. What was the input?

14. A function machine: input → ×6 → +4 → ÷2 → output. What is the output when the input is 5? A) 15 B) 16 C) 17 D) 18 E) 19

15. A function machine: input → ×(5/3) → −1 → output. What is the output when the input is 6?

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